General Mathematics — Exercise Sheet 39

Triple integrals and moments

Variant: Statements

Exercise 1 (A conical mass distribution). Let $K = \{(x, y, z) : 0 \le \sqrt{x^2 + y^2} \le z \le 2\}$ have density $\rho = z$. Use cylindrical coordinates to find its mass, centroid and moment of inertia about the z axis. Check the compact-domain hypotheses of Lecture 39. Use Lecture 26 for the resulting proper polynomial integrals.

Exercise 2 (An ellipsoid and an offset axis). Let $a, b, c, \rho_0 > 0$ and let the ellipsoid $K = \{x^2/a^2 + y^2/b^2 + z^2/c^2 \le 1\}$ have uniform density ρ_0. Use a linear change from the unit ball to compute its mass, centroid and moment I_z. Then find the moment about the line $(d, 0, t)$, $t \in \mathbb{R}$, where $d \in \mathbb{R}$. Use Lecture 39's ball calculation and parallel-axis theorem, and Lecture 22's determinant rule.

Exercise 3 (Mass concentrating near the boundary). A ball of radius $R > 0$ has radial density $\rho = \kappa r^p$, where $\kappa > 0$ and $p \ge 0$. At the origin set $\rho = \kappa$ if $p = 0$ and $\rho = 0$ if $p > 0$. Find its mass and moment about any central axis. Express $I/(mR^2)$ as a function of p and find its limit as $p \to \infty$. Use Lecture 39's spherical integration and reflection/permutation argument.

Exercise 4 (Negative radial moments). For $a > 0$, use the normalised density $w_a(r) = 4a^{3/2} r^2 e^{-ar^2}/\sqrt{\pi}$ on $r \ge 0$. Extend Lecture 39's moment formula to all real powers k, identifying exactly when the ordinary integral exists. Compute the moments for $k = -1$ and $k = -2$, and compute the variance $\int_0^\infty (r - \bar{r})^2 w_a(r)\,\mathrm{d}r$, where $\bar{r}$ is the mean. Use Lecture 27's endpoint comparison, Gamma integral and monotone cutoffs, and Lecture 26's integral mean value theorem.

Exercise 5 (The Beta–Gamma relation, sketch). For $p, q > 0$ define the ordinary improper integral

$$B(p, q) = \int_0^1 t^{p-1}(1 - t)^{q-1}\,\mathrm{d}t.$$

Show convergence and give a compact-cutoff sketch [S] of $B(p, q) = \Gamma(p)\Gamma(q)/\Gamma(p + q)$. Use only proper compact changes of variables from Lecture 38, ordinary endpoint limits from Lecture 27 and integral order. Evaluate $B(1/2, 3/2)$ and derive $B(p+1, q) = pB(p, q)/(p+q)$, retaining the sketch status of this route.

Exercise 6 (Laplace transforms and their exact domains). Define $\mathcal{L}[f](s) = \int_0^\infty e^{-st} f(t)\,\mathrm{d}t$ as an ordinary improper integral, with real s. Compute the transforms and their complete real convergence domains for $f(t) = t^n$ $(n = 0, 1, 2, \ldots)$, $f(t) = e^{at}$ $(a \in \mathbb{R})$, and $f(t) = \sin(\omega t)$ $(\omega \in \mathbb{R})$. Include every zero-parameter and boundary case. Use Lectures 26–27, the Gamma recurrence and the trigonometric primitives.